COM205 Microelectronic Devices and Circuits

Quiz 1 Questions | Session 2025-04

Question 1

What is the percent regulation and load regulation of a voltage regulator? What are the minimum and maximum allowed values of variable input voltages for a Zener diode-based voltage regulator with a fixed load resistance R_L? [2+5=7m]

Question 2

For the common-emitter bipolar junction transistor (BJT) amplifier circuit in the Figure, where DC biasing voltage Vcc is 10 V, input source V_s has an RMS voltage amplitude of 7.07 mV, the source resistance R_s = 500 Ω, R_1 = 47 kΩ, R_2 = 10 kΩ, R_C = 5 kΩ, $R_{E1} = R_{E2}$ = 330 Ω, R_L = 56 kΩ, $C_1 = C_3$ = 10 μF, and C_2 = 100 μF. Assume that at the source frequency, the capacitors act like short circuits. (a) Determine the DC collector voltage. (b) Draw the load line and determine the Q point of this amplifier. Also, find out the maximum allowed base current I_b fluctuations for the reliable amplification of Vs and (c) Determine the peak amplitude of the AC collector voltage. [2+3+3=8m]

